

MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 12: TVD Schemes, Flux Limiters, and SSP Time Stepping

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1 Introduction: From Recipes to Theory

In Lecture 11 we cured the oscillations of Lax-Wendroff and Beam-Warming by *limiting* the slope of a piecewise linear reconstruction. The limiters (minmod, MC, superbee) were motivated by a picture: do not let the reconstruction create new extrema. The numerical results were convincing, but several questions were left open:

- What exactly does “non-oscillatory” mean, as a mathematical property of a scheme?
- Which limiters are guaranteed to be non-oscillatory, and which are not? Is there a limit on how compressive a limiter may be?
- The time integrator also matters. Why SSP-RK2 and not, say, classical RK4?
- What is the price of all this? Is the limited scheme really second order everywhere?

This lecture begins **Module 4: High-Resolution Methods for Conservation Laws** and answers these questions with the theory of **Total Variation Diminishing (TVD)** schemes, developed by Harten (1983) and Sweby (1984). Our roadmap:

1. Compare three notions of “non-oscillatory” (monotone, TVD, monotonicity preserving) and see why TVD is the right one for high-resolution schemes.
2. Prove **Harten’s lemma**, a simple sufficient condition for TVD.
3. Write limited schemes in **flux-limiter** form, and derive **Sweby’s TVD region** for the limiter function $\phi(r)$.
4. Compare the classical limiters in theory and in practice.
5. Show that **SSP Runge-Kutta** methods preserve the TVD property in time.
6. Identify the fundamental limitation of TVD schemes: loss of accuracy at smooth extrema.

2 What Does “Non-Oscillatory” Mean?

Throughout, $\bar{u}^n = \{\bar{u}_i^n\}_i$ denotes the grid function at time t^n , and we use the jumps $\Delta_{i+1/2}^n \equiv \bar{u}_{i+1}^n - \bar{u}_i^n$. Recall from Lecture 10 the total variation

$$\text{TV}(\bar{u}^n) = \sum_i \left| \Delta_{i+1/2}^n \right| = \sum_i \left| \bar{u}_{i+1}^n - \bar{u}_i^n \right|. \quad (1)$$

The exact entropy solution of a scalar conservation law never increases its total variation: shocks and rarefactions can destroy variation, but nothing can create it. A numerical scheme should respect this.

Three Notions of Non-Oscillatory Schemes

A conservative scheme $\bar{u}^{n+1} = \mathcal{H}(\bar{u}^n)$ is called

- (i) **monotone** if $\mathcal{H}_i$ is a non-decreasing function of every argument $\bar{u}_j^n$ (Lecture 10);
- (ii) **total variation diminishing (TVD)** if $\text{TV}(\bar{u}^{n+1}) \leq \text{TV}(\bar{u}^n)$ for all data;
- (iii) **monotonicity preserving** if monotone (say, non-increasing) data $\bar{u}^n$ always produce monotone data $\bar{u}^{n+1}$ of the same kind.

These notions are nested:

$$\text{monotone} \implies \text{TVD} \implies \text{monotonicity preserving}.$$

The first implication was stated in Lecture 10. The second is easy to see. Suppose $\bar{u}^n$ is monotone, with far-field values $u_{-\infty}$ and $u_{+\infty}$. For monotone data the total variation is as small as possible: $\text{TV}(\bar{u}^n) = |u_{+\infty} - u_{-\infty}|$. A conservative scheme does not change the far-field values, so if $\bar{u}^{n+1}$ had an interior extremum, the path from $u_{-\infty}$ to $u_{+\infty}$ would have to go up and down, and $\text{TV}(\bar{u}^{n+1}) > |u_{+\infty} - u_{-\infty}| = \text{TV}(\bar{u}^n)$, contradicting TVD.

Why not simply ask for monotone schemes? Because of the following result, which strengthens Godunov's Barrier Theorem (Lecture 8) to *nonlinear* schemes:

Theorem (Harten, Hyman and Lax, 1976)

A monotone conservative scheme for a scalar conservation law is at most **first-order accurate**.

So monotonicity is too strong: no amount of nonlinear cleverness can make a monotone scheme second order. TVD is weaker. It does not allow new extrema to form from monotone data, yet (as we will see) it admits schemes that are second order in smooth regions. This is why high-resolution schemes are built around the TVD property.

3 Harten's Lemma: A Practical Test for TVD

Checking $\text{TV}(\bar{u}^{n+1}) \leq \text{TV}(\bar{u}^n)$ directly for all data is hard. Harten found a condition that can be read off from the formula of the scheme.

Lemma (Harten, 1983)

Suppose a scheme can be written in the **incremental form**

$$\bar{u}_i^{n+1} = \bar{u}_i^n - C_{i-1/2} \Delta_{i-1/2}^n + D_{i+1/2} \Delta_{i+1/2}^n \quad (2)$$

where the coefficients $C_{i-1/2}$ and $D_{i+1/2}$ may depend on the data. If, for every i ,

$$C_{i+1/2} \geq 0, \quad D_{i+1/2} \geq 0, \quad C_{i+1/2} + D_{i+1/2} \leq 1, \quad (3)$$

then the scheme is TVD.

Proof. Subtract equation (2) at i from the same equation at $i+1$:

$$\begin{aligned} \Delta_{i+1/2}^{n+1} &= \Delta_{i+1/2}^n - C_{i+1/2} \Delta_{i+1/2}^n + C_{i-1/2} \Delta_{i-1/2}^n + D_{i+3/2} \Delta_{i+3/2}^n - D_{i+1/2} \Delta_{i+1/2}^n \\ &= (1 - C_{i+1/2} - D_{i+1/2}) \Delta_{i+1/2}^n + C_{i-1/2} \Delta_{i-1/2}^n + D_{i+3/2} \Delta_{i+3/2}^n, \end{aligned} \quad (4)$$

where all Δ on the right are at time t^n . By (3), all three coefficients are non-negative, so the triangle inequality gives

$$|\Delta_{i+1/2}^{n+1}| \leq (1 - C_{i+1/2} - D_{i+1/2}) |\Delta_{i+1/2}^n| + C_{i-1/2} |\Delta_{i-1/2}^n| + D_{i+3/2} |\Delta_{i+3/2}^n|. \quad (5)$$

Sum over all i and shift the index in the last two sums ($i - 1 \rightarrow i$ and $i + 1 \rightarrow i$):

$$\text{TV}(\bar{u}^{n+1}) \leq \sum_i (1 - C_{i+1/2} - D_{i+1/2} + C_{i+1/2} + D_{i+1/2}) |\Delta_{i+1/2}| = \text{TV}(\bar{u}^n). \quad (6)$$

3.1 Two Examples

1. **First-order upwind** for $u_t + au_x = 0$, $a > 0$: $\bar{u}_i^{n+1} = \bar{u}_i - \nu \Delta_{i-1/2}$. Here $C_{i-1/2} = \nu$ and $D_{i+1/2} = 0$, so the scheme is TVD exactly when $0 \leq \nu \leq 1$, the familiar CFL condition.
2. **Lax-Wendroff**: from Lecture 11,

$$\bar{u}_i^{n+1} = \bar{u}_i - \frac{\nu}{2} (\Delta_{i+1/2} + \Delta_{i-1/2}) + \frac{\nu^2}{2} (\Delta_{i+1/2} - \Delta_{i-1/2}), \quad (7)$$

so $C_{i-1/2} = \frac{\nu}{2}(1 + \nu)$ and $D_{i+1/2} = -\frac{\nu}{2}(1 - \nu)$. For $0 < \nu < 1$ we get $D < 0$: Harten's condition fails, consistent with the oscillations we observed. (Harten's condition is only sufficient, but here one can check directly that a single step on step data creates an overshoot, Exercise E2.)

4 Flux-Limiter Schemes

4.1 Blending a Low-Order and a High-Order Flux

A second way to describe limited schemes, due to Sweby (1984), works directly with fluxes. Take a monotone, first-order flux F^{low} and a second-order flux F^{high} , and blend them:

$$F_{i+1/2} = F_{i+1/2}^{\text{low}} + \phi(r_i) (F_{i+1/2}^{\text{high}} - F_{i+1/2}^{\text{low}}) \quad (8)$$

The **limiter function** ϕ depends on the **smoothness ratio**

$$r_i = \frac{\Delta_{i-1/2}}{\Delta_{i+1/2}} = \frac{\bar{u}_i - \bar{u}_{i-1}}{\bar{u}_{i+1} - \bar{u}_i}, \quad (9)$$

which compares the upwind jump to the local jump (for $a > 0$). Its meaning:

- $r_i \approx 1$: the data are smooth and nearly linear. Use the high-order flux ($\phi \approx 1$).
- $r_i \leq 0$: $\bar{u}_i$ is a local extremum. Use the low-order flux ($\phi = 0$).
- $r_i \gg 1$ or $0 < r_i \ll 1$: a steep gradient on one side, as near a discontinuity. Limit.

4.2 Linear Advection

For $u_t + au_x = 0$, $a > 0$, take F^{low} upwind and F^{high} Lax-Wendroff, which in the REA form of Lecture 11 (downwind slope) is $F_{i+1/2}^{\text{LW}} = a\bar{u}_i + \frac{a}{2}(1 - \nu)\Delta_{i+1/2}$. Then

$$F_{i+1/2} = a\bar{u}_i + \frac{a}{2}(1 - \nu)\phi(r_i)\Delta_{i+1/2} \quad (10)$$

Comparing with the REA flux $F_{i+1/2} = a\bar{u}_i + \frac{a}{2}(1 - \nu)\Delta x \sigma_i$ of Lecture 11, the flux-limiter scheme is the same thing as a slope-limiter scheme with

$$\Delta x \sigma_i = \phi(r_i) \Delta_{i+1/2} \quad (11)$$

So the two viewpoints are equivalent: every slope limiter is a flux limiter, and vice versa. The classical linear schemes correspond to simple choices of ϕ :

Scheme	$\phi(r)$	Slope $\Delta x \sigma_i$	Order
Upwind	0	0	1
Lax-Wendroff	1	$\Delta_{i+1/2}$	2
Beam-Warming	r	$\Delta_{i-1/2}$	2
Fromm	$\frac{1}{2}(1+r)$	$\frac{1}{2}(\Delta_{i-1/2} + \Delta_{i+1/2})$	2

And the minmod slope of Lecture 11 corresponds to $\phi(r) = \max(0, \min(1, r))$, because

$$\text{minmod}(\Delta_{i-1/2}, \Delta_{i+1/2}) = \Delta_{i+1/2} \text{minmod}(r_i, 1) = \Delta_{i+1/2} \max(0, \min(1, r_i)).$$

5 Sweby's TVD Region

5.1 Derivation

Substituting the flux (10) into the conservative update gives

$$\bar{u}_i^{n+1} = \bar{u}_i - \nu \Delta_{i-1/2} - \frac{\nu(1-\nu)}{2} [\phi(r_i) \Delta_{i+1/2} - \phi(r_{i-1}) \Delta_{i-1/2}]. \quad (12)$$

To bring this into Harten's incremental form, the key step is to rewrite the downwind jump using the definition of r_i : $\Delta_{i+1/2} = \Delta_{i-1/2}/r_i$. Then everything is a multiple of $\Delta_{i-1/2}$:

$$\bar{u}_i^{n+1} = \bar{u}_i - \nu \underbrace{\left[1 + \frac{1-\nu}{2} \left(\frac{\phi(r_i)}{r_i} - \phi(r_{i-1}) \right) \right]}_{C_{i-1/2}} \Delta_{i-1/2}, \quad D_{i+1/2} = 0. \quad (13)$$

Harten's conditions reduce to $0 \leq C_{i-1/2} \leq 1$. Writing $\Theta \equiv \phi(r_i)/r_i - \phi(r_{i-1})$, this is

$$-\frac{2}{1-\nu} \leq \Theta \leq \frac{2}{\nu}. \quad (14)$$

For $0 \leq \nu \leq 1$ both bounds are at least 2 in magnitude, so it is enough to ask $|\Theta| \leq 2$, and this holds for *any* pair r_i, r_{i-1} if

$$0 \leq \frac{\phi(r)}{r} \leq 2 \quad \text{and} \quad 0 \leq \phi(r) \leq 2. \quad (15)$$

For $r \leq 0$ the first condition forces $\phi(r) = 0$: at an extremum the scheme must drop to first order. We have proved:

Theorem (Sweby, 1984): TVD Region for Flux Limiters

The flux-limiter scheme (10) is TVD for $0 \leq \nu \leq 1$ provided

$$\boxed{\phi(r) = 0 \text{ for } r \leq 0, \quad 0 \leq \phi(r) \leq \min(2r, 2) \text{ for } r > 0.} \quad (16)$$

This region is shaded in Figure 1(a). The picture immediately explains Lecture 11: Lax-Wendroff ($\phi = 1$) leaves the region for $r < 1/2$, Beam-Warming ($\phi = r$) for $r > 2$, and Fromm for small and large r . Each linear second-order scheme is a straight line through $(1, 1)$, and no straight line through $(1, 1)$ other than $\phi \equiv 0$ can stay inside the region for all r . This is Godunov's barrier drawn as a picture. A nonlinear limiter bends the curve to stay inside.

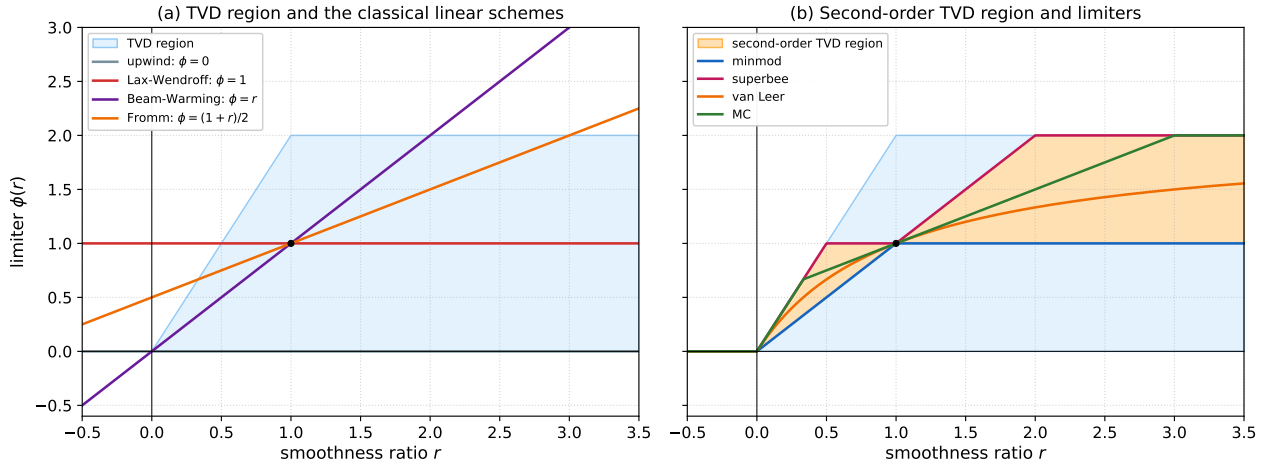


Figure 1: Sweby diagram. (a) The TVD region (16) (blue) and the classical linear schemes, all straight lines through $(1, 1)$; each leaves the TVD region. (b) The second-order TVD region (orange), bounded below by minmod and above by superbee, with the van Leer and MC limiters in between.

5.2 Second-Order Accuracy

Inside the TVD region we still want second order in smooth regions. There $r_i = 1 + \mathcal{O}(\Delta x)$, and one can show that the scheme is second-order accurate if $\phi(1) = 1$ and ϕ is smooth (Lipschitz) near $r = 1$. Sweby further recommended that ϕ should lie between the Beam-Warming line $\phi = r$ and the Lax-Wendroff line $\phi = 1$, so that the scheme is locally a weighted average of two second-order schemes. Intersected with (16), this gives the **second-order TVD region** of Figure 1(b):

$$\min(r, 1) \leq \phi(r) \leq \max(\min(2r, 1), \min(r, 2)), \quad r > 0. \quad (17)$$

The lower boundary is the **minmod** limiter and the upper boundary is the **superbee** limiter: they are the two extreme second-order TVD limiters.

5.3 Symmetry

A natural further requirement is that the limiter treats forward and backward gradients alike, so that a profile and its mirror image are limited in the same way. In terms of ϕ this is

$$\frac{\phi(r)}{r} = \phi\left(\frac{1}{r}\right). \quad (18)$$

All four limiters below satisfy it (Exercise E3).

6 The Classical Limiters

Limiter	$\phi(r)$	Character
Minmod (Roe, 1986)	$\max(0, \min(1, r))$	Lower edge of the second-order region. Most diffusive, very robust.
Superbee (Roe, 1985)	$\max(0, \min(2r, 1), \min(r, 2))$	Upper edge. Most compressive: very sharp discontinuities, but steepens smooth waves into squares.
van Leer (1974)	$\frac{r + r }{1 + r }$	Smooth (differentiable for $r > 0$), a good compromise.
MC (van Leer, 1977)	$\max(0, \min(2r, \frac{1}{2}(1 + r), 2))$	Follows Fromm near $r = 1$. Sharp and accurate, a popular default.

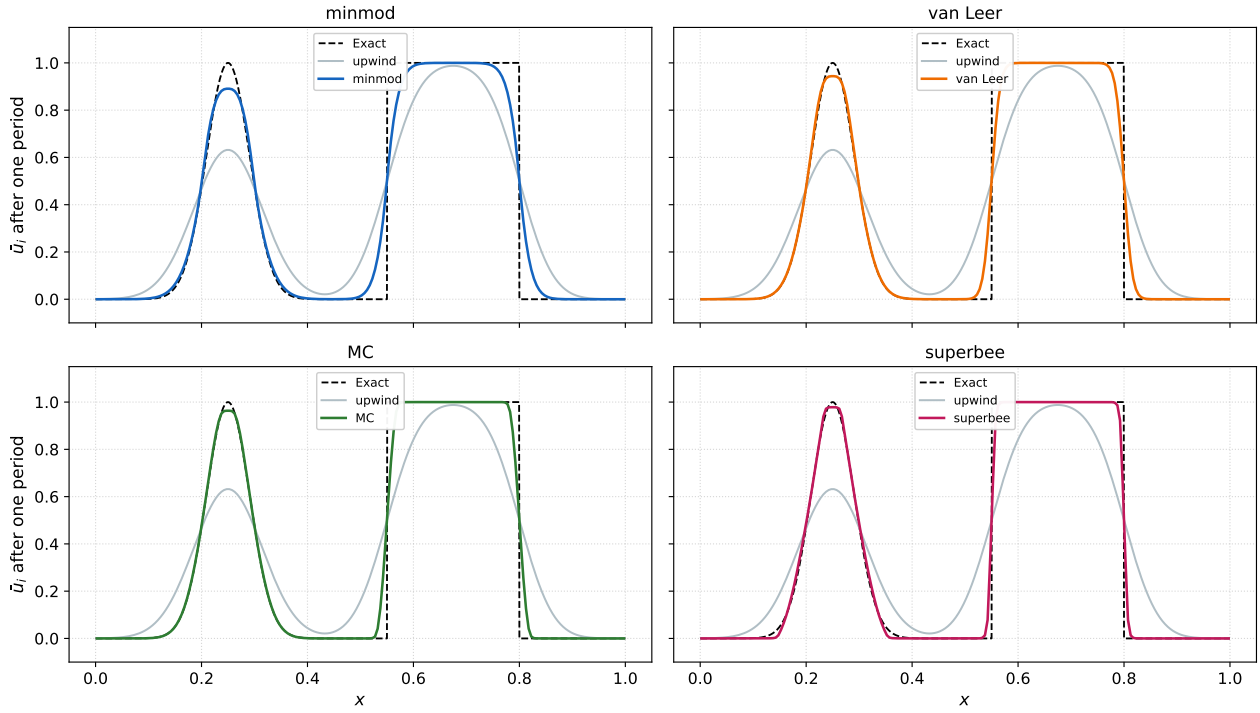


Figure 2: Flux-limiter schemes (10) for $u_t + u_x = 0$ with a Gaussian and a square wave, periodic domain, $N = 200$, $\nu = 0.5$, after one period. First-order upwind (grey) is shown for comparison in every panel.

Figure 2 advects a profile containing both a smooth Gaussian and a square wave through one full period, the kind of test that exposes the character of each limiter:

- **Minmod** is non-oscillatory but visibly diffusive: the square wave edges are rounded and the Gaussian peak is cut down noticeably.
- **van Leer** and **MC** keep the edges sharp and preserve most of the Gaussian peak.
- **Superbee** gives the sharpest square wave of all, but it starts to over-steepen the Gaussian, squaring off its top and steepening its sides.

There is no universally best limiter. A diffusive limiter is safe but smears; a compressive limiter is sharp but distorts smooth waves. In practice MC and van Leer are the most common general-purpose choices.

7 The Semi-Discrete MUSCL Scheme is TVD

Lecture 11 used the Method of Lines: limited interface states in space, then SSP-RK2 in time. Let us check the TVD property for its building block, a single forward Euler step. For $a > 0$ with the upwind flux, $F_{i+1/2} = a u_{i+1/2}^L = a(\bar{u}_i + \frac{1}{2}\Delta x \sigma_i)$, and with $\Delta x \sigma_i = \phi(r_i)\Delta_{i+1/2}$:

$$\bar{u}_i^{n+1} = \bar{u}_i - \nu \left[\Delta_{i-1/2} + \frac{1}{2} (\phi(r_i)\Delta_{i+1/2} - \phi(r_{i-1})\Delta_{i-1/2}) \right] = \bar{u}_i - \nu \underbrace{\left[1 + \frac{1}{2} \left(\frac{\phi(r_i)}{r_i} - \phi(r_{i-1}) \right) \right]}_{C_{i-1/2}} \Delta_{i-1/2}. \quad (19)$$

This is the Sweby computation again, except that the factor $(1 - \nu)$ is missing, because forward Euler does not include the time correction built into the REA flux. If ϕ lies in the TVD region (16), the bracket lies in $[0, 2]$, so $0 \leq C_{i-1/2} \leq 2\nu$, and:

Result: TVD Time Step for Semi-Discrete MUSCL

Semi-discrete MUSCL with any limiter in the TVD region, advanced by one forward Euler step, is TVD for

$$\nu = \frac{a\Delta t}{\Delta x} \leq \frac{1}{2}. \quad (20)$$

This is the CFL number $1/2$ quoted in Lecture 11. (For minmod alone the bracket lies in $[\frac{1}{2}, \frac{3}{2}]$, which allows $\nu \leq 2/3$; see Exercise E4.) But forward Euler is only first order in time. We now show how to go to higher order in time without losing this property.

8 TVD in Time: Strong Stability Preserving Runge-Kutta Methods

8.1 The Problem with Ordinary Runge-Kutta Methods

In Lecture 9 we studied Runge-Kutta methods through their linear stability regions. Linear stability controls the growth of errors, but it says nothing about oscillations: a scheme can be linearly stable and still create a new overshoot at every step. Gottlieb and Shu showed examples where a TVD spatial operator combined with a perfectly stable, second-order Runge-Kutta method produces oscillations near shocks. We need a time integrator that *inherits* the TVD property of forward Euler.

8.2 The Shu-Osher Idea

Suppose the spatial operator $\mathcal{L}$ is such that one forward Euler step is TVD:

$$\text{TV}(\bar{u} + \Delta t \mathcal{L}(\bar{u})) \leq \text{TV}(\bar{u}) \quad \text{for } \Delta t \leq \Delta t_{\text{FE}}. \quad (21)$$

Shu and Osher (1988) observed that if every stage of a Runge-Kutta method is a *convex combination* of forward Euler steps, then TVD is automatically preserved, because the total variation is a seminorm and satisfies $\text{TV}(\alpha v + \beta w) \leq \alpha \text{TV}(v) + \beta \text{TV}(w)$ for $\alpha, \beta \geq 0$. Methods of this type are called **strong stability preserving (SSP)**, or TVD Runge-Kutta methods.

SSP Runge-Kutta Methods (Shu and Osher, 1988)

SSP-RK2 (second order):

$$\begin{aligned} \bar{u}^{(1)} &= \bar{u}^n + \Delta t \mathcal{L}(\bar{u}^n), \\ \bar{u}^{n+1} &= \frac{1}{2} \bar{u}^n + \frac{1}{2} [\bar{u}^{(1)} + \Delta t \mathcal{L}(\bar{u}^{(1)})]. \end{aligned} \quad (22)$$

SSP-RK3 (third order):

$$\begin{aligned} \bar{u}^{(1)} &= \bar{u}^n + \Delta t \mathcal{L}(\bar{u}^n), \\ \bar{u}^{(2)} &= \frac{3}{4} \bar{u}^n + \frac{1}{4} [\bar{u}^{(1)} + \Delta t \mathcal{L}(\bar{u}^{(1)})], \\ \bar{u}^{n+1} &= \frac{1}{3} \bar{u}^n + \frac{2}{3} [\bar{u}^{(2)} + \Delta t \mathcal{L}(\bar{u}^{(2)})]. \end{aligned} \quad (23)$$

Both are TVD under the *same* time step restriction $\Delta t \leq \Delta t_{\text{FE}}$ as forward Euler.

Proof for SSP-RK2. The first stage is a forward Euler step, so $\text{TV}(\bar{u}^{(1)}) \leq \text{TV}(\bar{u}^n)$ by (21). The bracket in the second stage is a forward Euler step from $\bar{u}^{(1)}$, so its total variation is at most $\text{TV}(\bar{u}^{(1)})$. Using the seminorm property,

$$\text{TV}(\bar{u}^{n+1}) \leq \frac{1}{2} \text{TV}(\bar{u}^n) + \frac{1}{2} \text{TV}(\bar{u}^{(1)} + \Delta t \mathcal{L}(\bar{u}^{(1)})) \leq \frac{1}{2} \text{TV}(\bar{u}^n) + \frac{1}{2} \text{TV}(\bar{u}^{(1)}) \leq \text{TV}(\bar{u}^n). \quad (24)$$

□

The same argument works stage by stage for SSP-RK3 (Exercise E5). Combined with Section 7: *semi-discrete MUSCL with a TVD limiter and SSP-RK2 or SSP-RK3 is TVD for $\nu \leq 1/2$* . SSP-RK3 is the standard time integrator for the high-order ENO and WENO schemes of the next lecture.

Insight: Two Kinds of Stability

Linear stability (Lectures 7 and 9) asks whether errors grow; it is analysed with Fourier modes and stability regions. **Strong stability** (this lecture) asks whether a nonlinear quantity such as the total variation can grow; it is analysed with convexity. Shock-capturing needs both, and SSP methods are designed for the second.

9 The Price of TVD: Clipping at Extrema

Every TVD limiter satisfies $\phi(r) = 0$ for $r \leq 0$. At a smooth maximum, such as the crest of a sine wave, the jumps on the two sides have opposite signs, so $r \leq 0$ and the scheme uses the first-order upwind flux there. The peak is therefore eroded a little at every time step. This effect is called **clipping**, and it is visible in the minmod panel of Figure 2.

The table below measures it. We advect $u_0(x) = \sin(2\pi x)$ for one period with $\nu = 0.5$ and compare the numerical cell averages with the exact ones.

Scheme	L^1 error			L^∞ error		
	$N = 320$	$N = 640$	order	$N = 320$	$N = 640$	order
Lax-Wendroff ($\phi = 1$)	1.93×10^{-4}	4.82×10^{-5}	2.00	3.03×10^{-4}	7.57×10^{-5}	2.00
Minmod	5.16×10^{-4}	1.37×10^{-4}	1.92	4.60×10^{-3}	1.86×10^{-3}	1.31
van Leer	1.19×10^{-4}	2.62×10^{-5}	2.19	1.89×10^{-3}	6.96×10^{-4}	1.44
MC	4.87×10^{-5}	9.44×10^{-6}	2.37	1.03×10^{-3}	3.47×10^{-4}	1.57

Two observations:

- In the L^1 **norm** the limited schemes are essentially second order. The clipped region is only a few cells wide, and it shrinks as the mesh is refined.
- In the L^∞ **norm** the order drops to between 1.3 and 1.6: the largest error sits exactly at the extrema, where the scheme is locally first order. Unlimited Lax-Wendroff has no such loss, but it oscillates at discontinuities.

This is not a defect of a particular limiter; it is a theorem.

Limitations of TVD Schemes

- (1) (**Osher and Chakravarthy, 1984**) In one dimension, a TVD scheme reduces to first-order accuracy at smooth extrema of the solution.
- (2) (**Goodman and LeVeque, 1985**) In two dimensions, a conservative scheme that is TVD is at most first-order accurate.

The second result is striking: the TVD concept, so successful in 1D, cannot be the foundation of truly high-order multidimensional schemes. The way out is to relax the requirement. Instead of demanding that the total variation never grows, we allow it to grow by a tiny amount that vanishes as $\Delta x \rightarrow 0$ (TVB schemes, Shu 1987), or we only demand that the scheme is *essentially* non-oscillatory: spurious oscillations, if any, are of the size of the truncation error. This is the idea behind **ENO and WENO schemes**, the subject of Lecture 13. They reconstruct with higher-degree polynomials and choose the stencil adaptively, rather than limiting the slope.

10 Summary

Concept	Statement	Significance
Hierarchy	monotone $\Rightarrow$ TVD $\Rightarrow$ monotonicity preserving	Monotone schemes are first order (Harten-Hyman-Lax); TVD allows second order
Harten's lemma	$C_{i+1/2}, D_{i+1/2} \geq 0, \quad C_{i+1/2} + D_{i+1/2} \leq 1 \Rightarrow$ TVD	A test that can be read off from the scheme
Flux limiter	$F = F^{\text{low}} + \phi(r)(F^{\text{high}} - F^{\text{low}})$	Equivalent to slope limiting: $\Delta x \sigma_i = \phi(r_i) \Delta_{i+1/2}$
Sweby region	$\phi = 0$ for $r \leq 0$; $0 \leq \phi \leq \min(2r, 2)$	TVD for $\nu \leq 1$; second order if also $\phi(1) = 1$
SSP-RK	Convex combinations of forward Euler steps	TVD in time with the forward Euler step size
Clipping	$\phi = 0$ at extrema	First order at smooth extrema; motivates ENO/WENO

11 Exercises

Exercises for Lecture 12

E1. Harten's Lemma for Lax-Friedrichs: The Lax-Friedrichs scheme for $u_t + au_x = 0$ is $\bar{u}_i^{n+1} = \frac{1}{2}(\bar{u}_{i+1} + \bar{u}_{i-1}) - \frac{\nu}{2}(\bar{u}_{i+1} - \bar{u}_{i-1})$.

- (a) Write it in incremental form and identify $C_{i-1/2}$ and $D_{i+1/2}$.
- (b) Show that it is TVD for $|\nu| \leq 1$, including negative a .

E2. Lax-Wendroff is not TVD: Take $\bar{u}_i = 1$ for $i \leq 0$ and $\bar{u}_i = 0$ for $i \geq 1$, and $\nu = 1/2$.

- (a) Perform one Lax-Wendroff step and compute $\bar{u}_0^{n+1}$ and $\bar{u}_1^{n+1}$.
- (b) Show that an overshoot appears and compute $\text{TV}(\bar{u}^{n+1})$.
- (c) Repeat with the minmod flux limiter and verify that TV does not increase.

E3. Limiter Identities:

- (a) Using (11), show that the MC and superbee slopes of Lecture 11 correspond to the $\phi(r)$ in the table of classical limiters.
- (b) Show that the van Leer limiter corresponds to the harmonic-mean slope $\Delta x \sigma_i = \frac{2\Delta_{i-1/2}\Delta_{i+1/2}}{\Delta_{i-1/2} + \Delta_{i+1/2}}$ when the jumps have the same sign (and 0 otherwise).
- (c) Verify the symmetry property $\phi(r)/r = \phi(1/r)$ for minmod, superbee, van Leer and MC.
- (d) Show that each of the four limiters lies in the second-order TVD region and satisfies $\phi(1) = 1$.

E4. Time Step for Minmod MUSCL: For the semi-discrete scheme of Section 7 with the minmod limiter, show that $\frac{1}{2} \leq 1 + \frac{1}{2}(\phi(r_i)/r_i - \phi(r_{i-1})) \leq \frac{3}{2}$, and deduce that one forward Euler step is TVD for $\nu \leq 2/3$.

E5. SSP-RK3 Preserves TVD: Assuming (21), prove that each stage of SSP-RK3 satisfies $\text{TV}(\bar{u}^{(k)}) \leq \text{TV}(\bar{u}^n)$, and hence that $\text{TV}(\bar{u}^{n+1}) \leq \text{TV}(\bar{u}^n)$ for $\Delta t \leq \Delta t_{\text{FE}}$.

E6. Computational Exercise: Run `advection_flux_limiters.jl` from the course website.

- (a) Try all limiters on the Gaussian plus square wave profile and rank them by the height of the Gaussian after one period.
- (b) Replace the limiter by $\phi(r) = \frac{1}{2}(1 + r)$ (Fromm, not limited). Where do oscillations appear?
- (c) Design your own limiter inside the second-order TVD region, for example $\phi(r) = \max(0, \min(1.5r, 1), \min(r, 1.5))$, and compare it with minmod and superbee.

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